Persistent Information in Large Agent Populations

Scaling, Incentives, and Coordination

Research summary accompanying the integrated working paper

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The connected question

How do the scaling of information deposits and the incentives to produce them jointly determine the information environment that survives in a large population? Exogenous production establishes the possible limits; the game asks which one decentralized incentives sustain. A productive coordinated policy shows that disappearance can coexist with a strictly useful alternative. Creator credit examines one response and its limitations.

One common model

Agent i has state $Y_i = (X_i, C_i) \in \{0,1\} \times \{U, E, A\}$: a binary task coordinate and an unexposed, exposed, or adopter communication state. The population law is $\mu^N = N^{-1} \sum_i \delta_{Y_i}$, and $a(\mu)$ is its adopter share. A shared board $B \geq 0$ decays at rate κB. Communication follows

$$U \xrightarrow{\rho B} E, \qquad E \xrightarrow{\alpha} A, \qquad E \xrightarrow{\nu} U, \qquad A \xrightarrow{\delta} U.$$

Tasks complete at rate c_0 for nonadopters and c_1 for adopters, and reset at rate r. Deposits change only the board. The game observes labeled states and the board, and controls only artifact production. [Paper, §2]

Exogenous channel	Per-adopter rate	Board jump	Limiting effect
Routine	λ_w	η/N	Deterministic drift
Artifact	λ_M/N	h	State-thinned common jumps
Hybrid	Both channels	Both sizes	Drift plus common jumps

These rates are prescribed, not equilibrium choices. In the game, an eligible player's artifact rate is $\lambda_M u_i/N$, with $0 \leq u_i \leq \bar{u}$. Frozen unit production is feasible as a game policy only when $\bar{u} \geq 1$.

What the probability foundation establishes

Writing $Q(B)$ for the common transition matrix, the frozen hybrid limit is

$$\dot{\mu}_t = \mu_t Q(B_t), \qquad dB_t = [-\kappa B_t + \eta \lambda_w a(\mu_t)]dt + h\, dJ_t,$$

where J accepts a common Poisson proposal at height at most $\lambda_M a(\mu_{t-})$. The weak and rare limits disable the corresponding channel. The board jumps; the population law remains continuous.

On each fixed horizon, the expected uniform error in population, board, and accepted count is at most $C_T(\varepsilon_{N,p} + N^{-1/2})$ for these frozen models. Assumptions include bounded initial boards, L^p convergence to deterministic initial data for some $p > 2$, and independence from future clocks. The proof includes the random finite hybrid board explicitly. [Paper, Thms. 3.2–3.4; App. A]

With exchangeable chaotic initialization, weak-limit tagged paths have an ordinary product law. Rare and hybrid tags factor conditional on the common proposal measure. Averaging over that history gives a mixture of products, so unconditional dependence may remain. This is not finite-N independence.

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Incentives select a limiting environment

The allocation-and-readiness institution. Opportunities arrive at total rate $\lambda_M \bar{u}$ and are allocated uniformly among N players. An eligible recipient succeeds with probability $u_i/\bar{u}$. Ineligible assignments expire and cannot be reassigned. Players choose and pay for readiness before allocation, whether an opportunity arrives or not. Readiness units, cap, costs, and per-person rewards stay fixed across N. Fixed capacity alone does not imply these restrictions. [Paper, §5]

Only adopters choose effort, with cost $c(u) = \ell u + \gamma u^2/2$, $\ell \geq 0$, $\gamma > 0$. On $[0, T]$, players maximize expected discounted reward minus cost, with $0 \leq g \leq R$, $\beta \geq 0$, and zero terminal payoff. Welfare is the per-person average. There are initially no transfers.

An equilibrium exists before taking the limit. For every finite N, T, the paper constructs a pure continuous Markov-perfect equilibrium and verifies it against admissible predictable unilateral deviations. Every pure Borel Markov-perfect equilibrium has the same values and induced state/count law from every restart; arbitrary Borel strategy representatives need not coincide pointwise. The construction and Borel proof are included. [Paper, Thms. 5.1, 5.3; App. B]

Private incentives suppress artifacts. Every pure Borel state-feedback Nash equilibrium satisfies the following eligible-state relation almost everywhere along equilibrium play; continuous locally verified feedbacks satisfy it pointwise. Write $\Delta_h V_i = V_i(s^h) - V_i(s)$ and $K_T = R \int_0^T e^{-\beta t} dt$:

$$u_i^{\mathrm{NE}} = \mathrm{clip}_{[0,\bar{u}]} \left(\frac{(\lambda_M/N)\Delta_h V_i - \ell}{\gamma} \right) \leq \frac{[\lambda_M K_T/N - \ell]_+}{\gamma}.$$

Thus, for quadratic cost,

$$\Lambda_N^{\mathrm{NE}} = \frac{\lambda_M}{N} \sum_i \mathbf{1}_{\{C_i = A\}} u_i^{\mathrm{NE}} \leq \frac{\lambda_M^2 K_T}{\gamma N}.$$

Positive ℓ gives eventual zero effort cost and artifact count. Nash optimality yields the maximizing identity without a verification assumption; initial-law Nash gives the lower value bound only along equilibrium play. [Paper, Cor. 5.2; Thm. 5.3]

For these classes, equilibrium and artifact-disabled paths agree until the first artifact, whose fixed-horizon probability is $O(N^{-1})$. Their expected uniform state and welfare differences are $O(N^{-1})$ with bounded initial boards. Combining this with the weak foundation gives a deterministic equilibrium limit, possibly with routine drift, and ordinary tagged independence under deterministic limiting initialization and exchangeable chaotic tags. [Paper, Prop. 7.1]

Disappearance can coexist with productive coordination

Take rare-only production, $B_0 = 0$, reward $g(x, c) = Rx$, positive $R, \rho, \alpha, h, \lambda_M, \bar{u}$, and $c_1 > c_0$. Uniformly positive fractions initially are adopters and incomplete unexposed players. A sufficiently small fixed effort $\varepsilon > 0$ by adopters until $t_0 < T$ yields

$$W_N^\varepsilon - W_N^0 = \varepsilon A_N + O(\varepsilon^2), \quad A_N \geq A_* > 0, \qquad \liminf_N (W_N^{\mathrm{SP}} - W_N^{\mathrm{NE}}) \geq \delta_* > 0.$$

The welfare comparison uses these same equilibrium classes and this seeded, zero-board, rare-only region. Small positive linear costs can also be allowed; arbitrary initial boards and hybrid productivity require a separate positivity argument. [Paper, Thms. 6.1, 6.3]

The same specified policy has a common-jump limit. For two limiting tagged discounted task rewards H_i, conditional independence and a productive early-artifact event give $\mathrm{Cov}(H_1, H_2) = \mathrm{Var}(\mathbb{E}[H_1 \mid \Pi^0]) > 0$. This shows that the common history affects useful outcomes, not that randomness itself raises welfare. For general bounded spatially Lipschitz feedback, the paper gives a quantitative rare-only limit and a qualitative hybrid limit. Neither optimal-value nor optimal-control convergence is established. [Paper, Thm. 7.2; §7]

Two scaling questions, and a benchmark response

Physical deposition scaling. For an order-one eligible share, individual rate $\lambda_0 N^{-\zeta}$ and positive impact $\eta_0 N^{-\xi}$ give aggregate rate, drift, and predictable quadratic-variation orders

$$N^{1-\zeta}, \qquad N^{1-\zeta-\xi}, \qquad N^{1-\zeta-2\xi}.$$

For $\zeta, \xi \geq 0$, order-one positive drift lies on $\zeta+\xi = 1$. Positive ξ gives vanishing jumps and fluctuations; $(1, 0)$ is the finite-jump endpoint. The accepted rate can be state dependent. This is uncentered positive-deposit accounting, not a Brownian-limit theorem or an incentive classification. [Paper, Prop. 3.1]

Strategic productivity scaling. The two-state benchmark instead has absent/present artifact state Z, benefit vZ, depreciation rate d, infinite horizon, and fixed $\beta > 0$. When absent, discovery is $a_N \sum_i u_i$. With quadratic cost, $a_N = \lambda N^{-s}$, fixed positive $\lambda, v, \gamma, \bar{u}$, and $k = \beta + d$, the symmetric stationary equilibrium's absent-state hazard satisfies

$$q_N^{\mathrm{NE}} \sim \begin{cases} \lambda\sqrt{v/\gamma}\, N^{1/2-s}, & 0 \leq s < 1/2, \\ \lambda^2 D_c/\gamma, & s = 1/2, \\ \lambda^2 v(\gamma k)^{-1} N^{1-2s}, & s > 1/2, \end{cases} \qquad kD_c + \lambda^2 D_c^2/\gamma = v.$$

Every fixed positive cap is eventually nonbinding. Positive linear cost instead shuts down every $s > 0$ eventually. At $s = 0$, activity requires $\lambda v > k\ell$, with limiting hazard $\lambda v/\ell - k$. These are benchmark results, not six-state scaling theorems. Stationary event frequency is $dq_N/(d + q_N)$ when $d > 0$, rather than the conditional hazard q_N. No exchange of population and long-horizon limits is asserted. [Paper, Prop. 4.3; §4]

With normalized productivity $a_N = \lambda/N$, Nash artifact intensity vanishes but an active benchmark planner sustains a positive constant hazard. The capped and linear-cost cases, waiting-time laws, occupancy, and values are solved in the paper. Concentrating the same opportunity stream on one designated creator also gives an N-independent private problem. Capacity alone therefore cannot explain collapse. This comparison does not derive endogenous hierarchy or efficient delegation. [Paper, Prop. 4.2]

Creator credit: exact implementation and a counterexample

Pay the successful creator Np and charge every player p per discovery, including the creator. Transfers balance event by event. In the two-state benchmark, $p = D^{\mathrm{SP}}$, the planner's value gap, implements planner effort as a symmetric stationary Markov equilibrium. The assumptions include risk neutrality, reliable attribution, enforceable levies, and commitment. [Paper, Prop. 8.1]

For an interior six-state planner target, the actual individual continuation value under transfers, U_i^p, must instead satisfy

$$\Delta_h U_i^p + (N-1)p = N\Delta_h W_N^{\mathrm{SP}}.$$

Budget balance gives only an average value identity. Setting $p = \Delta_h W_N^{\mathrm{SP}}$ need not satisfy this individual condition. In a two-agent example, a completed adopter has $\Delta_h U_1^p = 0$; halving prescribed effort over a positive interval, then reverting, gives a strictly positive gain. This proves nonimplementation, not a full optimal dynamic best-response policy. [Paper, Prop. 8.2; §8]

Under active normalized benchmark implementation and positive expected discovery, a tagged player's gross creator payments tend to zero in probability but keep a positive expectation: they are not uniformly integrable. State-path convergence alone therefore cannot pass expected transfer payoffs to a limit. [Paper, §8]

Numerical evidence and a secondary application

The illustrated six-state policy uses eligible effort 0.25 until time 1.25 on a horizon of 6, with zero board, no routine deposits, 20% initial adopters, and all tasks incomplete. Each cell uses 160 independent histories; the paper specifies the full configuration and methods. [Paper, §10]

Population	Welfare above exact baseline (95% interval)
60	0.1147 $[0.0469, 0.1825]$
240	0.1947 $[0.1184, 0.2710]$
960	0.1240 $[0.0618, 0.1863]$
Fixed-policy limit	0.0998 $[0.0442, 0.1554]$

These are per-capita discounted net rewards with Student intervals; the exact baseline is 1.15065544. The conservative witness certifies effort about 3.70×10^{-7} and gain about 1.78×10^{-10}. Improvement at 0.25 is simulation evidence; the $0.25/N$ diagnostic is not a Nash solve. V3 retains v2's histories, aggregates, and numerical checks. The new theorem is proved analytically. Code remains local.

The weak limit also closes on exposed share, adopter share, and board stock. Its loop gain is

$$G_B = \frac{\rho\alpha\eta\lambda_w}{(\alpha+\nu)\delta\kappa}.$$

Under the stated positive-rate assumptions, $G_B = 1$ is a forward transcritical threshold with local full-simplex stability exchange. No global full-system stability is asserted. This secondary reservoir-style result is distinct from the production game. [Paper, Prop. C.1]

Interpretation and remaining boundaries

History-dependent and randomized equilibria obey the same sharp bound under explicit no-direct-effort-monitoring and implementable shadow-deviation assumptions. General action-monitored history equilibria remain unresolved. Singleton-time changes disprove pointwise Borel strategy uniqueness and shutdown, not artifact disappearance. [Paper, Thm. D.1; App. D]

The access institution is unvalidated for AI-agent systems. Bayesian learning, endogenous hierarchy, efficient specialization, general heterogeneous-state implementation, and optimizer convergence are not established. The scalar board omits content and artifact-specific exposure. Reassigning unused opportunities at fixed capacity and task dynamics remains a next question.

Related work

Team discovery and durable public goods exhibit incentive wedges [2, 1]; common-Poisson limits supply probability context [3]. No priority claim is made.

Selected references

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[2] Alessandro Bonatti and Johannes Hörner. Collaborating. *American Economic Review*, 101(2):632–663, 2011. doi: 10.1257/aer.101.2.632. URL https://bpb-us-w2.wpmucdn.com/campuspress.yale.edu/dist/3/352/files/2012/02/65916.pdf.

[3] Daniel Hernández-Hernández and Joshué Helí Ricalde-Guerrero. Conditional McKean–Vlasov differential equations with common Poissonian noise: Propagation of chaos. *Stochastics and Dynamics*, 25(7–8):2550033, 2025. doi: 10.1142/S0219493725500339. URL https://arxiv.org/abs/2308.11564. Article 2550033; author version arXiv:2308.11564v3.